Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 08.09.2026**

Assignment-5 (Image Enhancement in Frequency Domain)

1. The given grayscale image "gaussian_noise.png" is degraded by *additive white Gaussian noise* (AWGN) with $mean\,\mu = 0$ and $variance\,\sigma^2 = 0.01$, where the image intensities are normalized to the range $[0, 1]$. Reduce the effect of noise using frequency-domain low-pass filtering. Resize both "gaussian_noise.png" and its clean "reference.png" to 256×256 and implement the 2-D DFT of the noisy image to obtain its frequency-domain representation $F(u, v)$. Shift the zero-frequency component to the center and display the noisy image and its centered Fourier spectrum. Construct a *Gaussian Low-Pass Filter* (GLPF) with cutoff distance $D_0 = 30$ pixels using

$$H(u, v) = e^{[-D^2(u,v) / (2D_0{}^2)]}$$

where $D(u, v)$ is the Euclidean distance of the frequency coordinate (u, v) from the center of the spectrum. Display the constructed filter $H(u, v)$ and perform element-wise multiplication with the centered Fourier transform to obtain the filtered frequency-domain representation $G(u, v)$ as

$$G(u, v) = H(u, v)F(u, v)$$

Implement the 2-D inverse DFT and apply it to $G(u, v)$ to obtain the noise-reduced image $g(x, y)$ in the spatial domain and display the reconstructed image. Evaluate the filtering performance by calculating the *Peak Signal-to-Noise Ratio* (PSNR) of both the noisy image and the noise-reduced image with respect to the "clean reference.png".
Where,

$$MSE = \left(\frac{1}{MN}\right) \Sigma_{x,\ y} [R(x, y) - I(x, y)]^2$$
$$PSNR = 10\, log_{10}(1/MSE)$$

where $R(x, y)$ is the clean reference image, $I(x, y)$ is the image being evaluated, and $M \times N$ is the image size. Calculate and compare the MSE and PSNR values of both images. A higher PSNR value indicates that the image is closer to the clean reference and contains less distortion.

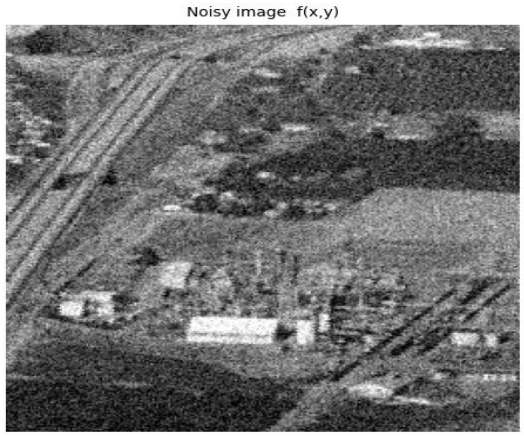
[20 Marks]

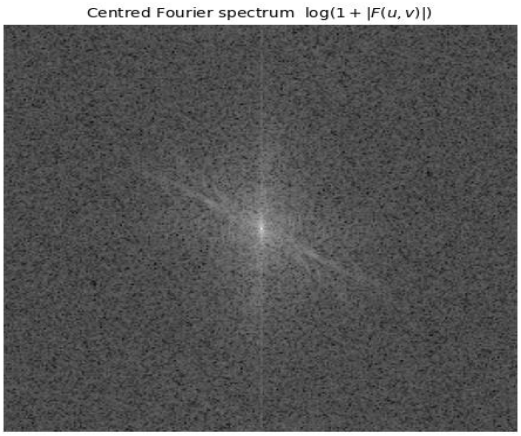
Sample Input:

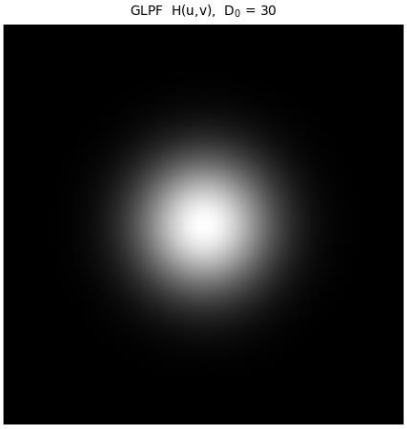




Sample Output:







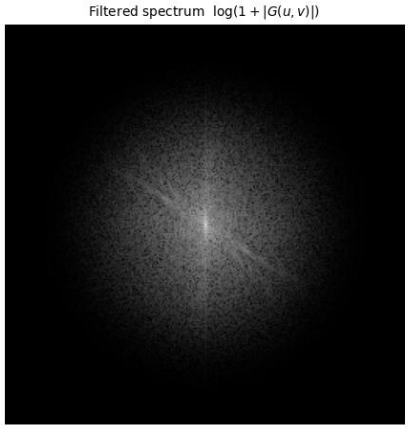




Image	MSE	PSNR (dB)
Noisy image	0.009796	20.090
Noise-reduced image	0.003178	24.978

2. The edges and fine structural details in the given grayscale image "building.png" are not sufficiently prominent. Enhance these details using frequency-domain high-pass filtering. Resize the input image to 256×256 and implement the 2-D DFT to obtain its frequency-domain representation $F(u, v)$. Shift the zero-frequency component to the center and display the original image and its centered Fourier spectrum. Construct a Butterworth High-Pass Filter (BHPF) of order $n = 2$ with cutoff distance $D_0 = 30$ pixels using

$$H(u, v) = 1 / \{1 + [D_0 / D(u, v)]^{2n}\}$$

where $D(u, v)$ is the Euclidean distance of the frequency coordinate (u, v) from the center of the spectrum. Display the constructed filter $H(u, v)$ and perform element-wise multiplication

with the centered Fourier transform to obtain the filtered frequency domain representation $G(u, v)$ as

$$G(u, v) = H(u, v)F(u, v)$$

Implement the 2-D inverse DFT and apply it to $G(u, v)$ to obtain the high-pass filtered image $g(x, y)$ in the spatial domain and display the reconstructed image. To evaluate the change in edge information, apply the standard 3×3 Sobel horizontal and vertical filters separately to both the original image $f(x, y)$ and the high-pass filtered image $g(x, y)$ using a manually implemented convolution. Compute the $Tenengrad$ (T), the sharpness measure for each image as

$$T = \left(\frac{1}{MN}\right) \Sigma_{x,y} \left[G_x{}^2(x, y) + G_y{}^2(x, y)\right]$$

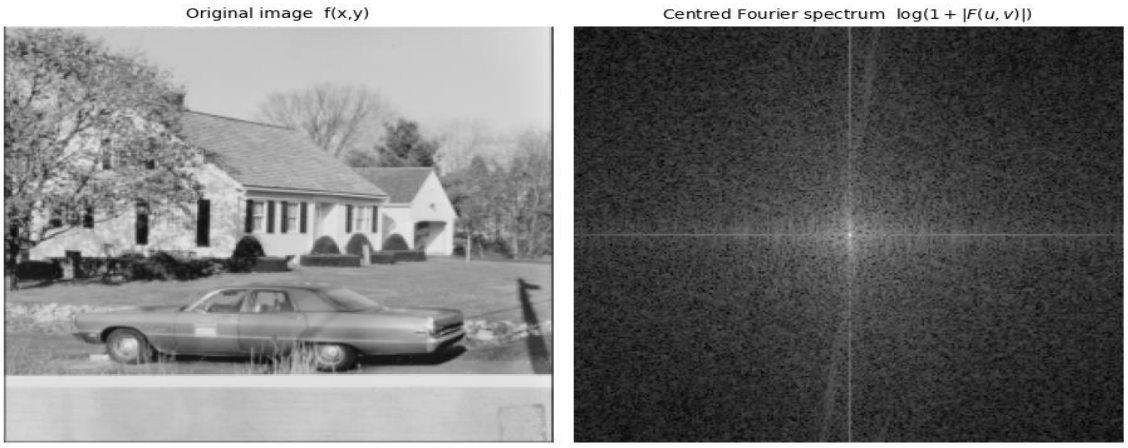
where G_x and G_y are the horizontal and vertical Sobel responses, respectively, and $M \times N$ is the image size. Calculate and compare the $Tenengrad$ values of both images. A higher $Tenengrad$ value indicates stronger edges and greater fine-detail content, while a lower value indicates weaker edge information.

[20 Marks]

Sample Input:



Sample Output:



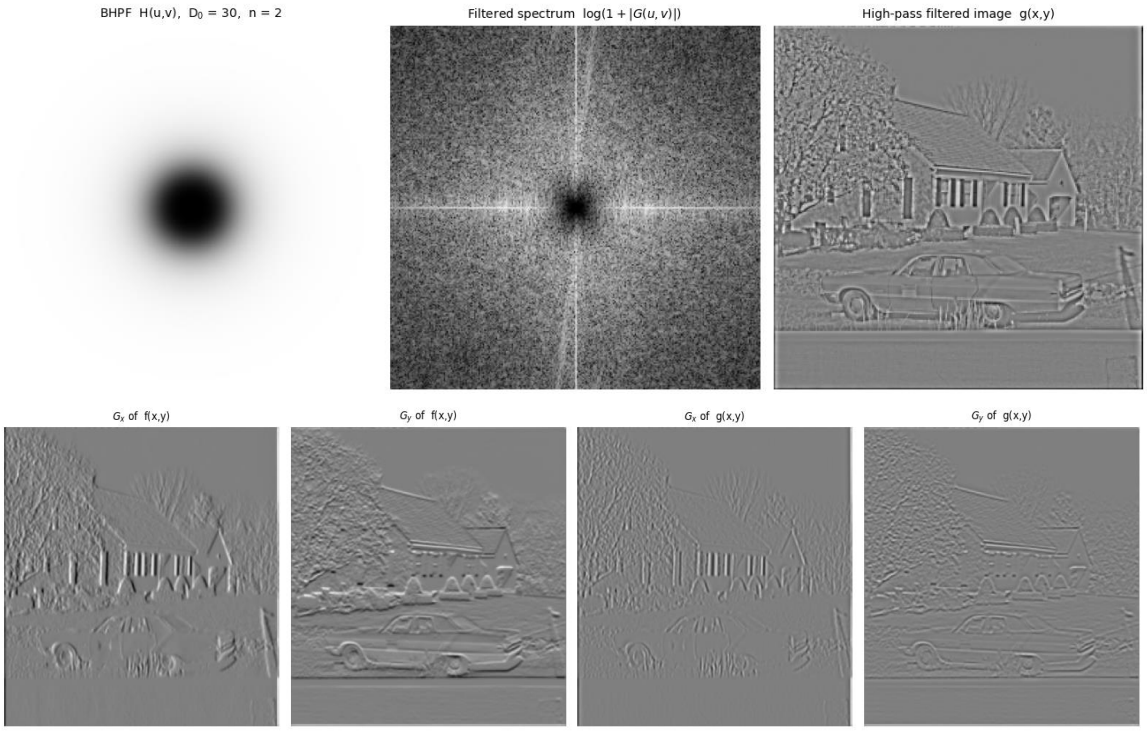


Image	Tenengrad (T)
Original image	0.30271
High-pass filtered	0.16252